

Linear Elasticity

April 21, 2026

REFERENCES

Ciarlet, *Mathematical Aspects of DG Methods*, SIAM, 2002.

Larson–Bengzon, *The Finite Element Method*, Springer, 2013.

Linear elasticity: setting

Let $\Omega \subset \mathbb{R}^3$ be the volume occupied by an elastic material. For every subdomain $\omega \subset \Omega$ with outward normal n :

- ▶ body forces: f ,
- ▶ boundary forces: $T_n := \sigma \cdot n$, **traction vector** with

$$(\sigma \cdot n)_i = \sum_{j=1}^3 \sigma_{ij} n_j$$

where σ is the stress tensor: the component σ_{ij} is the force per unit area in the x_i direction on a surface orthogonal to x_j .

By balance of angular momentum,

$$\sigma_{ij} = \sigma_{ji}.$$

The net force on ω is

$$F = \int_{\omega} f \, dx + \int_{\partial\omega} \sigma n \, ds = \int_{\omega} (f + \nabla \cdot \sigma) \, dx.$$

Cauchy equilibrium

From

$$F = \int_{\omega} f \, dx + \int_{\partial\omega} \sigma n \, ds = \int_{\omega} (f + \nabla \cdot \sigma) \, dx,$$

equilibrium for arbitrary ω gives the **Cauchy equilibrium equations**

$$-\nabla \cdot \sigma = f$$

In components,

$$-\sum_{j=1}^3 \partial_j \sigma_{ij}(u) = f_i.$$

These are 3 equations for the 6 stress components $\rightarrow$ requires *constitutive equations*.

Constitutive equations: relating stress to deformation

Introduce the displacement field

$$u = x - x_0,$$

where x_0 is the reference position.

Under the small-deformation assumption, the Green-Lagrange strain

$$E(u) = \frac{1}{2}(\nabla u + \nabla u^T + \nabla u^T \nabla u),$$

(measure of deformation) linearises to

$$\varepsilon(u) = \frac{1}{2}(\nabla u + \nabla u^T), \quad \varepsilon_{ij}(u) = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right).$$

This approximation is valid for **small strains** and **small rotations**.

We use a constitutive law to stress to deformation. **Hooke's law**:

$$\sigma = \mathbb{C} \varepsilon(u)$$

where $\mathbb{C}$ is a fourth-order elasticity tensor.

Lamé parameters and boundary conditions

Assuming the body is stress free before deformation, for linear elastic isotropic materials, Hooke's law becomes

$$\sigma(u) = 2\mu \varepsilon(u) + \lambda (\nabla \cdot u)I$$

In terms of Young's modulus E and Poisson ratio ν_P ,

$$\mu = \frac{E}{2(1 + \nu_P)}, \quad \lambda = \frac{E\nu_P}{(1 + \nu_P)(1 - 2\nu_P)}.$$

If the material is homogeneous, then λ, μ are constants.

Boundary conditions:

- ▶ Dirichlet: $u = g_D$ on Γ_D , eg. $u = 0$ for **clamped** boundaries
- ▶ Neumann: $\sigma n = g_N$ on Γ_N , Prescribing the **surface traction**, i.e. the normal stress.

with $\partial\Omega = \Gamma_D \cup \Gamma_N$ and $\Gamma_D \cap \Gamma_N = \emptyset$. We also assume $|\Gamma_D| > 0$; otherwise rigid body motions are not controlled

Strong form of linear elasticity

The elastostatic problem is

$$\begin{aligned} -\nabla \cdot \sigma &= f && \text{in } \Omega \\ \sigma(u) &= 2\mu \varepsilon(u) + \lambda(\nabla \cdot u)I && \text{in } \Omega \\ u &= g_D && \text{on } \Gamma_D \\ \sigma n &= g_N && \text{on } \Gamma_N \end{aligned}$$

Substituting Hooke's law gives

$$-\nabla(\lambda \nabla \cdot u) - \nabla \cdot (\mu \nabla u) - \nabla \cdot (\mu \nabla u^T) = f.$$

If λ, μ are constant,

$$-\mu \Delta u - (\lambda + \mu) \nabla(\nabla \cdot u) = f.$$

since $\nabla \cdot (\nabla u^T) = \nabla(\nabla \cdot u)$.

Weak formulation: first step

Assume $g_D = 0$. Let

$$V := \{v \in [H^1(\Omega)]^3 : v = 0 \text{ on } \Gamma_D\}.$$

Test the equilibrium equation with $v \in V$:

$$(f, v)_\Omega = (-\nabla \cdot \sigma, v)_\Omega.$$

Integrating by parts,

$$(f, v)_\Omega = \int_{\Omega} \sigma : \nabla v \, dx - \int_{\partial\Omega} (\sigma n) \cdot v \, ds = \int_{\Omega} \sigma : \nabla v \, dx - \int_{\Gamma_N} g_N \cdot v \, ds.$$

Symmetric gradient in the weak form

Decompose

$$\nabla v = \varepsilon(v) + \frac{1}{2}(\nabla v - \nabla v^T).$$

Since σ is symmetric and the skew-symmetric part of ∇v is orthogonal to every symmetric tensor,

$$\sigma : \nabla v = \sigma : \varepsilon(v).$$

Hence

$$\int_{\Omega} \sigma(u) : \varepsilon(v) \, dx = \int_{\Omega} f \cdot v \, dx + \int_{\Gamma_N} g_N \cdot v \, ds.$$

Substituting Hooke's law,

$$\int_{\Omega} (2\mu \varepsilon(u) : \varepsilon(v) + \lambda(\nabla \cdot u)(\nabla \cdot v)) \, dx = \ell(v),$$

where

$$\ell(v) := \int_{\Omega} f \cdot v \, dx + \int_{\Gamma_N} g_N \cdot v \, ds.$$

Variational problem

Define

$$a(u, v) := \int_{\Omega} (2\mu \varepsilon(u) : \varepsilon(v) + \lambda(\nabla \cdot u)(\nabla \cdot v)) \, dx.$$

Then the weak problem is:

Find $u \in V$ such that

$$a(u, v) = \ell(v) \quad \forall v \in V.$$

In components,

$$a(u, v) = \sum_{i,k,\ell} \int_{\Omega} (\lambda \partial_{\ell} u_{\ell} \partial_k v_k + \mu \partial_{\ell} u_k \partial_{\ell} v_k + \mu \partial_k u_{\ell} \partial_{\ell} v_k) \, dx.$$

Equivalently,

$$a(u, v) = \int_{\Omega} \left(\lambda(\nabla \cdot u)(\nabla \cdot v) + \mu \nabla u : \nabla v + \mu \nabla u^T : \nabla v \right) dx.$$

Well-posedness: Korn's inequality

Korn's inequality states that, for $V = [H_{\Gamma_D}^1(\Omega)]^3$ with $|\Gamma_D| > 0$,

$$\|v\|_{H^1(\Omega)} \leq C_K \|\varepsilon(v)\|_{L^2(\Omega)} \quad \forall v \in V.$$

In particular,

$$\|\nabla v\|_{L^2(\Omega)} \lesssim \|\varepsilon(v)\|_{L^2(\Omega)}.$$

If $\lambda \geq 0$ and $\mu \geq \mu_{\min} > 0$, then

$$a(v, v) = 2 \int_{\Omega} \mu |\varepsilon(v)|^2 dx + \int_{\Omega} \lambda |\nabla \cdot v|^2 dx \geq 2\mu_{\min} \|\varepsilon(v)\|_{L^2(\Omega)}^2,$$

so, by Korn,

$$a(v, v) \gtrsim \|v\|_{H^1(\Omega)}^2.$$

Finite element approximation

Approximate each component by a standard H^1 -conforming finite element space. For example,

$$V_h := \{v_h \in V : v_h|_K \in [\mathbb{P}_k(K)]^3 \ \forall K \in \mathcal{T}_h\}.$$

The discrete problem is:

Find $u_h \in V_h$ such that

$$a(u_h, v_h) = \ell(v_h) \quad \forall v_h \in V_h.$$

This is simply a vector-valued conforming FEM.

Céa and error estimate

Under the assumptions of the theorem in the notes, the problem is well posed and

$$\|u - u_h\|_{H^1(\Omega)} \leq C \inf_{v_h \in V_h} \|u - v_h\|_{H^1(\Omega)}.$$

Hence, if $u \in [H^{r+1}(\Omega)]^3$ and $r \leq k$,

$$\|u - u_h\|_{H^1(\Omega)} \leq C \left(\sum_{K \in \mathcal{T}_h} h_K^{2r} |u|_{H^{r+1}(K)}^2 \right)^{1/2} \lesssim h^r \|u\|_{H^{r+1}(\Omega)}.$$

In particular, for degree k elements and sufficient regularity, the method is $\mathcal{O}(h^k)$ in H^1 .